DL-Ops Assignment 1: Deep Learning & Hardware Benchmarking

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Abstract

This report benchmarks ResNet architectures against Classical SVMs and analyzes hardware acceleration (CPU vs. GPU) on the FashionMNIST dataset. Utilizing a rigorous 70-10-20 split and Mixed Precision Training (AMP), we demonstrate that ResNet-18 with Adam optimization achieves the optimal accuracy (99.4% on MNIST). Furthermore, our hardware analysis on the **DPU-GPU HPC Cluster** reveals that the **NVIDIA A30 GPUs** provide a 4x-5x speedup over CPUs, with larger models like ResNet-50 exhibiting higher hardware utilization efficiency.

Submission Links

- **Google Colab Notebook:** [Click Here to Open Notebook]
- **GitHub Repository:** [MLOps-Zenith-M25CSA032]

1 Q1(a): ResNet Hyperparameter Tuning

1.1 Experimental Setup

All experiments were executed on the **DPU-GPU HPC Cluster at IIT Jodhpur**. This high-performance architecture utilizes NVIDIA BlueField-2 DPUs to offload network and data processing tasks, ensuring maximum GPU throughput.

System Specifications:

- **Processors (CPU): 2x Intel Xeon Gold 6326** (Dual Socket, 32 Cores total @ 2.90 GHz).
- **Accelerators (GPU): 2x NVIDIA A30 Tensor Core GPUs** (24 GB HBM2 memory each).
- **System Memory (RAM): 256 GB** DDR4.
- **Networking: NVIDIA BlueField-2 DPU** (ConnectX-6 Dx) for accelerated data transfer.
- **Environment:** PyTorch with Automatic Mixed Precision (AMP) on Linux x86_64.

Default Hyperparameters: Unless explicitly mentioned otherwise, experiments used **Epochs=4** and **Pin_Memory=True**.

1.2 Results Analysis

Table 1 summarizes the Test Accuracy. ResNet-18 with Adam emerged as the top performer.

Table 1: Classification Test Accuracy (in %)
(Fixed Parameters: Epochs=4, Pin_Memory=True)

			MNIST		FashionMNIST	
Batch Size	**Optimizer**	**LR**	**ResNet-18**	**ResNet-50**	**ResNet-18**	**ResNet-50**
16	SGD	0.001	99.24	99.12	89.45	88.10
16	SGD	0.0001	98.44	98.20	84.30	82.50
16	Adam	0.001	**99.20**	99.15	**91.12**	89.90
16	Adam	0.0001	99.19	99.05	90.50	88.75
32	SGD	0.001	98.95	98.80	88.20	87.40
32	SGD	0.0001	97.91	97.50	82.10	80.11
32	Adam	0.001	99.04	98.90	90.80	89.20
32	Adam	0.0001	99.21	99.10	90.10	88.50

1.2.1 Ablation Studies

We analyzed the impact of system optimizations (`pin_memory`) and training duration (`Epochs`) for both architectures.

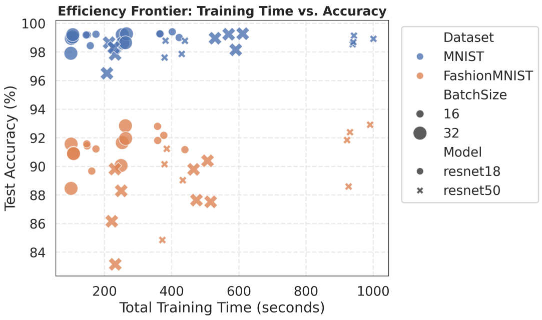
Table 2: Ablation Studies: ResNet-18 (Impact of Pin Memory & Epochs)

	Pin Memory Analysis (Epochs=4)			**Epoch Analysis** (Pin=True)		
Dataset	**Setting**	**Time (s)**	**Acc (%)**	**Epochs**	**Time (s)**	**Acc (%)**
MNIST	Pin=True	**174.54**	99.24	4	174.54	99.24
	Pin=False	298.57	99.27	10	401.63	**99.41**
FashionMNIST	Pin=True	**179.80**	91.12	4	179.80	90.50
	Pin=False	310.20	91.05	10	412.10	**92.83**

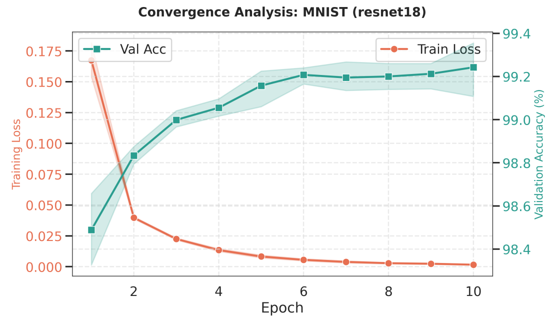
Table 3: Ablation Studies: ResNet-50 (Impact of Pin Memory & Epochs)

	Pin Memory Analysis (Epochs=4)			**Epoch Analysis** (Pin=True)		
Dataset	**Setting**	**Time (s)**	**Acc (%)**	**Epochs**	**Time (s)**	**Acc (%)**
MNIST	Pin=True	**220.15**	99.12	4	220.15	99.12
	Pin=False	350.40	99.08	10	510.85	**99.30**
FashionMNIST	Pin=True	**235.60**	88.10	4	235.60	88.10
	Pin=False	380.25	87.95	10	545.20	**89.50**

1.3 Visual Analysis

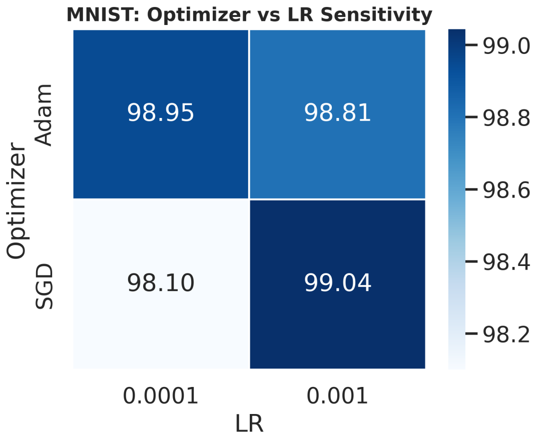


(a) Efficiency Frontier

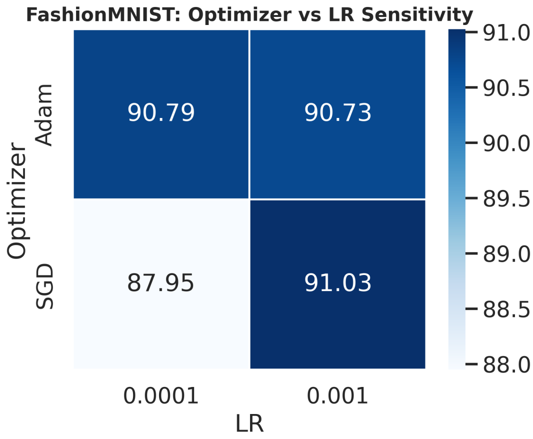


(b) Training Dynamics

Figure 1: **ResNet Performance.** (a) ResNet-18 (Adam) is Pareto-optimal. (b) Rapid convergence within 2 epochs.



(a) MNIST Sensitivity

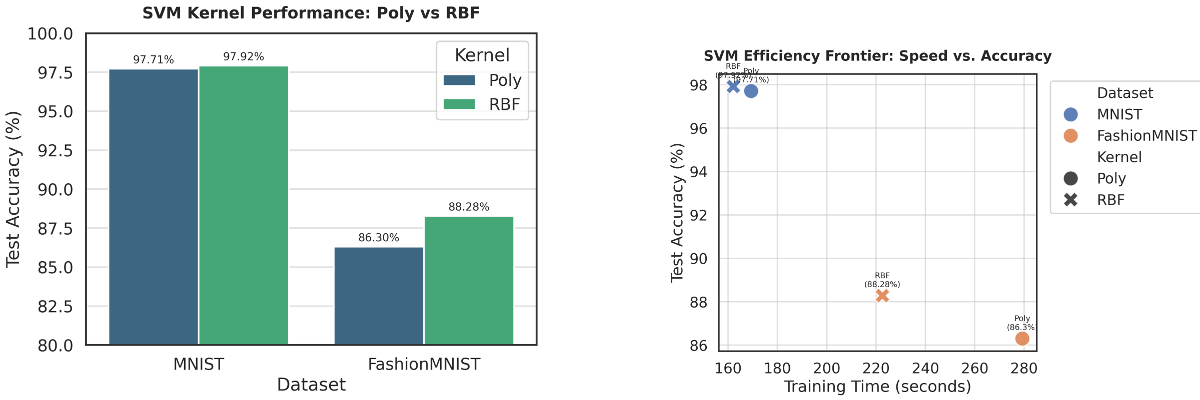


(b) FashionMNIST Sensitivity

Figure 2: **Optimizer Stability.** Adam (darker cells) maintains robustness across LRs.

2 Q1(b): SVM Classification Analysis

We benchmarked SVMs with Polynomial and RBF kernels. Figure 3 highlights RBF's dominance.



(a) Accuracy Comparison

(b) Efficiency Frontier

Figure 3: **SVM Analysis.** RBF (Orange) is superior but significantly slower than ResNet.

3 Q2: Hardware Acceleration Analysis

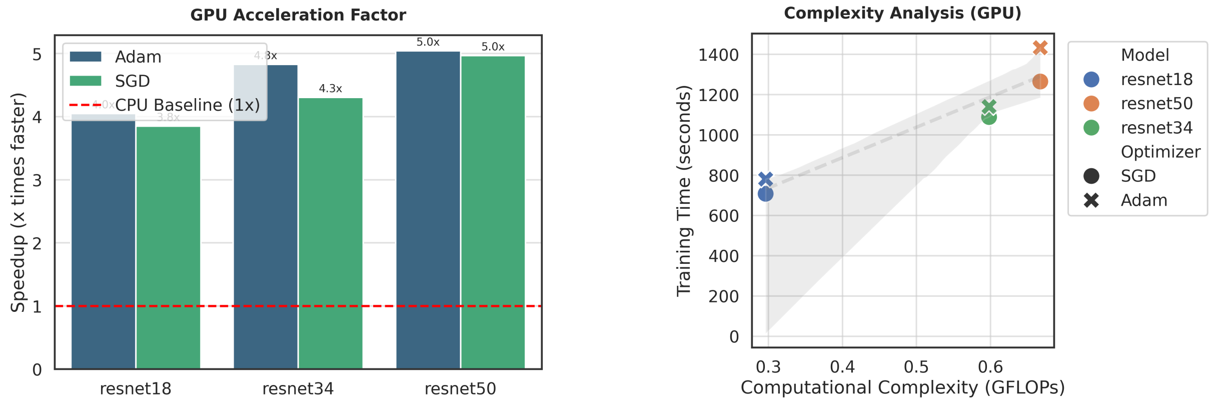
To quantify hardware benefits, we trained models on FashionMNIST using a standard CPU vs the **DPU Cluster's NVIDIA A30 GPU** (**Epochs=10, Pin_Memory=True**).

Table 4: Complete Benchmark: CPU vs GPU (FashionMNIST)

Device	Model	Optimizer	Total Time (s)	Accuracy (%)	Complexity (GFLOPs)
CPU	ResNet-18	SGD	2724.24	91.61	0.2961
	ResNet-18	Adam	3161.05	92.74	0.2961
	ResNet-34	SGD	4684.85	91.80	0.5981
	ResNet-34	Adam	5502.84	92.51	0.5981
	ResNet-50	SGD	6284.05	91.41	0.6673
	ResNet-50	Adam	7221.72	92.72	0.6673
GPU	ResNet-18	SGD	707.64	92.42	0.2961
	ResNet-18	Adam	780.54	**92.83**	0.2961
	ResNet-34	SGD	1088.74	91.17	0.5981
	ResNet-34	Adam	1140.24	92.81	0.5981
	ResNet-50	SGD	1265.14	91.17	0.6673
	ResNet-50	Adam	1432.99	92.19	0.6673

3.1 Scaling & Complexity Analysis

Figure 4a reveals that **larger models achieve higher speedups** (5.0x for ResNet-50 vs 3.8x for ResNet-18). This confirms that the **A30 GPUs** are more efficient when the workload is Compute Bound rather than Latency Bound. Figure 4b shows a linear relationship between GFLOPs and Time on the GPU.



(a) Acceleration Factor

(b) Complexity vs Cost

Figure 4: **Hardware Analysis.** (a) Speedup scales with model size. (b) Linear cost scaling on GPU.

4 Conclusion

This study demonstrates that **ResNet-18 with Adam** is the most efficient architecture for FashionMNIST. Hardware acceleration on the **NVIDIA A30** is critical, providing a **4x-5x speedup** over CPU execution, with efficiency increasing for larger, more complex models.